🎢 Theme Park · Petting Zoo — Helper Functions — Helper Functions 2 (Extension · Intermediate)

Topic: Petting Zoo — Helper Functions · **Course:** Theme Park Creator · **Level:** Intermediate · **Time:** about 50 minutes

This is your week 4 code. One **boss** function, three **helpers**, and every block placed **relative** to `x` and `z`.

```
def post(x, z):
    blocks.fill(OAK_FENCE, pos(x, 0, z), pos(x+15, 0, z+15), FillOperation.REPLACE)
    blocks.fill(AIR, pos(x+1, 0, z+1), pos(x+14, 0, z+14), FillOperation.REPLACE)

def tree(x, z):
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+3))
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+7))
    blocks.place(OAK_SAPLING, pos(x+3, 0, z+11))

def animals(x, z):
    for i in range(7):
        mobs.spawn(CHICKEN, pos(x+6, 0, z+3))

def petting_zoo(x, z):
    post(x, z)
    tree(x, z)
    animals(x, z)

petting_zoo(5, 5)
```

1 · Run Order 🔢

You type `petting_zoo(5, 5)`. The boss runs its lines **top to bottom**, and each helper finishes **completely** before the next starts.

a) Number the order these happen: fence built ⬜ · trees placed ⬜ · chickens spawned ⬜.

b) Could a chicken appear before the fence exists? Why / why not?

2 · Work Out the Coordinates 📐

With `petting_zoo(5, 5)`, `x = 5` and `z = 5`. Add the offsets and fill the real numbers.

Thing	Code	Real x	Real z
Fence far corner	`pos(x+15, 0, z+15)`		
First tree	`pos(x+3, 0, z+3)`		
Third tree	`pos(x+3, 0, z+11)`		
Chicken spawn	`pos(x+6, 0, z+3)`		

3 · Predict — Two Zoos 🔮

You add a second line: `petting_zoo(30, 5)`.

a) What are the two far corners of the second fence?

b) The first fence reaches x = 20. The second starts at x = 30. Do the two zoos touch, overlap, or leave a gap? Show the numbers.

> 💡 Because every block comes from `x` and `z`, moving a whole zoo is **one number**. That is the power of parameters.

4 · Spot the Bug 🐛

Bug A — All 7 chickens land in the same spot.

```
def animals(x, z):
    for i in range(7):
        mobs.spawn(CHICKEN, pos(x+6, 0, z+3))
```

Why? Rewrite the spawn line so each chicken is 1 block further along than the last.

Not a bug, but messy — `tree` is three copy-paste lines. The saplings sit at `z+3`, `z+7`, `z+11` (step of 4).

Rewrite `tree` using a `for` loop so the three saplings come from *one* line inside the loop.

5 · Extend the Zoo 💧

Add a `pond` helper, and call it from the boss. Then add **one more helper of your own** (a gate, a bench, a path) and call that too.

```
def pond(x, z):
    blocks.fill(WATER, pos(x+8, 0, z+9), pos(x+11, 0, z+12), FillOperation.REPLACE)
```

Write your new helper and your updated `petting_zoo` boss (with all the calls):

6 · Show Your Work 🎥

Record **one video**, one take, on your phone. Show in order:

1 · Your code. Scroll through it. Point at the boss and every helper.

2 · Your zoo. Point the camera. Name the fence, trees, animals, pond, and your own helper.

Then say these out loud, filling the blanks:

> The boss function is __. **It calls these helpers:** ____.
>
> I moved / resized the zoo by changing _____.
>
> The trickiest part was __ **because** ____.

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